

A Spectrum of Models for Low-Level Feedback Control Design of Small Underactuated Uncrewed Surface Vehicles

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Abstract—Effective low-level feedback control of small, underactuated uncrewed surface vehicles (USVs) — vessels under seven meters with a single thruster and rudder — requires dynamic models that provide actionable design insight, not predictive accuracy alone. Tow-tank testing is impractical at this scale; parameters must be identified from limited field trials. Existing system identification literature largely pursues model fidelity as an end in itself, producing representations whose complexity can obscure which aspects of the dynamics are structurally significant for compensator design.

We propose a methodology built around a deliberate spectrum of three model levels, each identified from the same experimental data: (1) first-order, linear, single-input single-output (SISO) transfer functions — Nomoto-style models for surge and yaw that are immediately interpretable and support controller architecture intuition, but risk oversimplification when considered in isolation; (2) a physics-based nonlinear state-space formulation incorporating quadratic hydrodynamic drag; and (3) a Nonlinear AutoRegressive with eXogenous inputs (NARX) multilayer perceptron. Rather than seeking a single best model, the spectrum is used collectively: simple models anchor human intuition; structured nonlinear models expose which terms are dynamically salient; and the black-box model bounds the benefit of added complexity.

Preliminary results from a step-response trial demonstrate the approach. First-order SISO models achieve $R^2=0.93$ (surge, $\tau=1.95$ s) and $R^2=0.83$ (yaw rate, $\tau=0.47$ s), providing direct controller tuning parameters. Cross-model diagnostics identify quadratic drag as the dominant surge nonlinearity and reveal that speed-dependent rudder effectiveness — a key coupling for underactuated control design — is not resolvable from a single short trial, motivating targeted data collection. The NARX model matches one-step prediction accuracy ($R^2=0.93$) but diverges in closed-loop simulation, demonstrating that physical structure provides essential regularization under limited data. A comprehensive dataset spanning multiple operating regimes is being collected to evaluate the full methodology.

Index Terms—uncrewed surface vehicle, system identification, model hierarchy, feedback control design, nonlinear dynamics, data-driven modeling, underactuated vessels

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